

Paper Creation Process Record (English Version)

Paper: *When Same-Prompt LLM Source Diversity Doesn't Help: An Ablation of Semantic Mutation Operators in Metamorphic Testing for Single-Output Scientific Computing Kernels* **Author:** Meng Li (University of South China) **Target venue:** *Information and Software Technology* (IST), Elsevier **Record date:** 2026-05-02 **Collaborator AI:** Claude Opus 4.7 (1M context) **Generated by:** Stage 6 PROCESS SUMMARY of the academic-pipeline orchestrator

1. Collaboration timeline

Date	Stage	Key milestone	Commit
2026-04-29 / before	Stage 1 RESEARCH	P1 12-PUT infrastructure + P2 three-stage ablation data (v3/v3b/v4) in place	(P1 repo)
2026-04-29 / before	Stage 2 WRITE	Chinese draft □□□□P2.md complete (83 KB; §1–§9 + abstract + keywords)	pre-b62db60
2026-04-30	Stage 3 REVIEW (R1)	5-reviewer simulated review (R0 EIC + R1 Method + R2 Domain + R3 Persp + R4 Devil's Advocate); verdict Major Revision ; 5/5 unanimous on 4 CRITICAL items	docs/review_2026-05-01/
2026-04-30 to 2026-05-01	Stage 4 REVISE (R1)	9 ≥ 3/5 reviewer-consensus items closed (P0-1 title scope / P0-2 17.6:1 removed / P0-3 sign test 4/4 demoted / P0-4 permutation null + Bonferroni / P0-5 chained conditioning / P0-6 IST 2024 reference withdrawn / P1-3 §9 L1-L6 / P1-5 zero-mass / P1-7 protocol asymmetry)	464779b..9a58c21
2026-05-01	R2 framework restructure	T1-T6 three-layer methodology backbone (§3.2.0 necessary conditions / §3.2.6.0 systematic vs incidental / §3.2 meta-operators + specialization / §2.3+§4.4 equivalence judgment Layer 2 / §3.2.6.3 mutant traceability / §1.2+§6+abstract narrative)	7847dea..9542b0f

Date	Stage	Key milestone	Commit
2026-05-02	NEW-MAJOR-1	12-PUT cosmic-ray full ablation: P2 =292, CR =1250, overall AST overlap 5.14% , HP/SI/TF=0/0/0 categorically unreachable	2547b61, b2d9b87
2026-05-02	R-1 (P0 blocker)	Full English translation □□□□P2_EN.md (BLTCY proxy + Anthropic Opus 4.7 streaming; 1844 lines, 150 KB); abstract + §1.2 sync in Chinese	37fa9bb
2026-05-02	Stage 3' RE-REVIEW (R2)	5-reviewer parallel re-review; verdict R0/R1/R2/R4 → Minor Revision , R3 → Major Revision round-2 §6.5-focused (D-6 deployability stays 3/10)	docs/review_2026-05-02/
2026-05-02	Stage 4' RE-REVISE	Group A (ESCALATED fixes: §3.5.1 EN missing / abstract H2 self-contradiction / §3.2.6.1 OS table cell / LLM source counts / §5.9.2-3 missing / line 1185 dangling) + Group B (R1 method: stipulated power 0.491 / Bonferroni × 4 Friedman / K_eq downgrade) + Group C (R2 lit: CPH grounding + 4 classics + Vargha-Delaney + Ammann-Offutt + §1.6.2 toy-scope + §9.5 Corollary) + Group D (R3 §6.5: thresholds removed / YAML excised + quarterly audit / air-gap declaration / §1.1 scope tightening / ASME V&V 20-2009 reference)	a20e795, a0fb8ed
2026-05-02	R3 Round-3 verification	R3 dissent withdrawn (D-6 raised to 6/10); 5/5 unanimous Minor Revision verdict	e7fe7d2

Date	Stage	Key milestone	Commit
2026-05-02	Stage 4.5 FINAL INTEGRITY	5-phase + 7-mode AI failure checklist; round-1 BLOCK (3 P0: Tip 2024 author / DeepCrime fully fabricated / cosmic-ray total 1276→1250); after fixes round-2 PASS_WITH_NITS (2 cosmetic)	1a5edef, bfac96f
2026-05-02	Stage 5 FINALIZE	Submission package: elsarticle LaTeX + xelatex PDF (124 pages / 380 KB / 0 missing-glyph) + Pandoc DOCX	c7333f0
2026-05-02	submit-prep	Author block populated (Meng Li, mlemon@usc.edu.cn, USC-CN) + IST cover letter generated	9ced0ad
2026-05-02	Stage 6 PROCESS SUMMARY	This document + Self-Reflection Report	this commit

Total collaboration duration: ~4 days (since 2026-04-29). **Total commits:** 30+ (full audit trail). **Final manuscript:** Chinese authoritative 1853 lines / English submission 1844 lines / 124-page PDF.

2. Collaboration quality assessment (6 dimensions, 1–100, honesty-first, no inflation)

Commitment: This assessment follows academic-pipeline §6 IRON RULE 7 — *no inflated scores*. Every score has specific evidence; numbers are not raised to avoid awkwardness.

2.1 Final manuscript quality — 78 / 100

Strengths (specific evidence): - §9 SMS→MS degeneration theorem (3 joint conditions L1/L2/L3 + 3 lemmas + main theorem + corollary) is a formalized, verifiable contribution (R2 Domain re-review verified □). - §3.2.6.3 12-PUT cosmic-ray ablation provides positive empirical refutation of the “new-concept classification” challenge (94.86% AST-disjoint; HP/SI/TF three classes categorically unreachable). - The three-layer methodology backbone (Layers 1/2/3) received substantive endorsement in R2 round-2 reviews (R0 7.4 / R2 7.43 / R4 5/5 CRITICAL → all neutralized). - §3.5.1 c-class shift handled honestly (selection-on-the-response declared explicitly + cross-cell exchangeability permutation null + Bonferroni × 5) — textbook post-hoc disclosure.

Weaknesses (specific evidence): - The H2 large-effect threshold ($\delta \geq 0.474$) was **not met** (primary $v3 = 0.323$; exploratory $v3b/v4 = 0.446/0.439$); the headline finding is “not met” rather than “achieved”. - §6.5 deployability remains a long-term aspiration with no traceable mapping to current IEC/ISO/ASME normative bodies (R3 D-6 stays at 6/10, not ≥8). - $n=12$ PUTs are toy-scope kernels; industrial-scale multi-module software empirics deferred to P5. - §6.5.3 in round-1 included numeric acceptance thresholds (0.20/0.30) which were removed in round-2 — a residue of originally over-claimed framing.

Why 78: The paper is a publication-grade IST submission, but not a top-tier benchmark. The three-layer methodology backbone is above the IST median methodological contribution, but the empirical scale and deployability claims fall short of industrial-scale work.

2.2 Methodological honesty — 88 / 100

Strengths: - Clean separation of pre-registered v3 vs exploratory v3b/v4 (§3.5.1 four caveats + permutation null + Bonferroni). - §3.2.6.3 OS row honestly revised from round-1 “□ tool-inexpressible” to “□ 88.33% disjoint + 11.67% incidental hits” — does not retroactively rescue the original strong claim for narrative consistency. - §5.7.3 stipulated-alternative power simulation (0.491) shows that “even at H2 boundary, sample randomness still produces failures ~50% of replications” — directly supports the H2 verdict’s point-estimate framing (P0-8 alignment). - §7 limitations cover R8-R12 residual threats; §7.5 K_eq-sweep promise honestly downgraded. - Withdrew the IST 2024 fabricated reference (commit ae609f1) and demoted 4 in-text citations to Tip-only + estimand caveat.

Weaknesses: - Stage 4.5 round-1 still found 3 P0 fabrication / drift issues (Tip 2024 mis-attributed authors / DeepCrime entry fully fabricated / cosmic-ray total 1276 vs 1250 drift) — these *AI-collaboration-induced* hallucinations were caught only by the final integrity check, indicating that in-conversation citation discipline was insufficient. - §1.3.2 had 3 orphan citations (Pradel/Cito/Tian) until round-2 cleanup.

Why 88: Methodological honesty is above the IST average, but AI-induced citation hallucinations (Modes 1 + 3) appearing in pre-final-check versions show that the review trail is a necessary safety net rather than redundant.

2.3 Collaboration efficiency — 75 / 100

Strengths: - Subagent-driven mode worked: round-1 revisions 9 items / R2 framework 6 items / round-2 Group A-D combined 24 items, executed largely in parallel. - Token efficiency: warning suppression + anchor-based grep + lean plan templates yielded ~30-50% per-call reduction. - Backed-up rationale: nearly every commit includes a W#/D#/CRITICAL# trace to a reviewer report.

Weaknesses: - Stage 5 LaTeX/PDF compile required 4 retries (elsarticle missing → CTAN download → newunicodechar missing → makecell missing → unicode-glyph postprocessing) — toolchain discovery was iterative cost. - R1 Methodology re-review subagent had a transient API timeout — required monitor + recovery coordination. - Stage 4.5 round-1 BLOCK + round-2 reverify was necessary, but the round-1 fabricated citations should have been caught earlier in the in-conversation phase.

Why 75: Collaboration was mostly smooth, but toolchain (LaTeX submission + subagent recovery) and citation discipline still produced ~25% non-productive time.

2.4 Feedback integration quality — 85 / 100

Strengths: - 5/5 reviewer consensus + DA CRITICAL items 100% addressed (no silent drops). - R&R Traceability (round-2 editorial-decision §2 verdict tally + §3 consensus + §4 disagreement matrix) closed item-by-item. - R3 Major-Revision dissent overcome via Group D §6.5 rewrite + Round-3 verification withdrawal — demonstrates iterative convergence of feedback integration.

Weaknesses: - Round-1 Group D arbitration items from R3 §6.5 were not fully executed in round-1 (numeric threshold removal / quarterly audit reframe were round-2 only) — round-1 editor’s P0 vs P1 classification leaned slightly lenient.

Why 85: Feedback integration is a strong dimension of this collaboration; both trace-ability and closure rate exceed the IST median.

2.5 Reproducibility & auditability — 92 / 100

Strengths: - 30+ commits with full git history; every commit traces to reviewer item. - All scripts (scripts/) + data (data/results/*.json SSOT) + mutant pools (data/mutants/*_pool_v4/) version-pinned. - REPRODUCIBILITY.md §4 explicitly documents the SMS_VERSION=v4 + P2_PRIMARY_VERSION=v3b dual-env-var convention (avoiding c-class drift). - docs/review_2026-05-01/ + docs/review_2026-05-02/ archive all reviewer reports for independent audit.

Weaknesses: - Minor data SSOT inconsistency between rq3_friedman_v4.json and paper_numbers_v4.json — does not affect manuscript content but housekeeping pending.

Why 92: Reproducibility is the strongest dimension of this collaboration; an independent reviewer should be able to reproduce all headline numbers within hours.

2.6 Self-correction capacity — 80 / 100

Strengths: - Stage 4.5 round-1 BLOCK → fix P0-A/B/C → round-2 PASS_WITH_NITS: **3 P0 issues caught and fixed by independent verifier before submission.** - R3 Major-Revision dissent → Group D 6 §6.5 fixes → Round-3 verification withdrew dissent: dissent not ignored but substantively processed. - §3.2.6.1 OS row self-revised from round-1 categorical claim to round-2 empirical “88.33%/11.67%” — does not retroactively rescue the error.

Weaknesses: - Tip 2024 author hallucination + DeepCrime fabricated entry **could have been avoided in-conversation** (no cite-then-verify discipline in round-1); Stage 2.5 INTEGRITY was not formally run in this collaboration (only implicitly via reviewer feedback) — Stage 4.5 was the first systematic execution. - AI-induced citation hallucination rate ~1-2% (15+ references → 2 confirmed-fabricated + 4 P1 orphans) — still above human-expert level.

Why 80: Self-correction is strong (BLOCK-triggered fix mechanism works), but root-cause prevention of AI hallucinations (cite-then-verify upfront) is weaker than post-hoc audit.

Overall weighted score

Equal-weight aggregation per academic-pipeline §6 template: $(78 + 88 + 75 + 85 + 92 + 80) / 6 = 83.0 / 100$

Tier: Above Average (above IST submission median, but not top-tier benchmark).

3. AI Self-Reflection Report (7-mode AI Research Failure Checklist Audit Log)

Per academic-pipeline v3.2 IRON RULE, Stage 2.5 + Stage 4.5 must run the 7-mode AI failure checklist. This section is the final audit log.

Mode	Failure mode	Stage 4.5 round-1	Stage 4.5 round-2	Resolution
1	Citation hallucination	SUSPECTED → CONFIRMED P0-A (Tip 2024 wrong authors) + P0-B (DeepCrime entry fully fabricated) + P1 orphans (Pradel/Cito/Tian)	PASS all citations resolve via WebFetch / Crossref	Fixed in commits 1a5edef + bfac96f
2	Implementation bug	PASS — script implementations match paper descriptions	PASS	—
3	Hallucinated results	SUSPECTED → CONFIRMED P0-C (cosmic-ray total 1276 vs SSOT 1250 drift) + P1 (Abstract +89% vs actual +91.4%; Spearman p version cross-talk)	PASS all numbers trace to SSOT JSON	Fixed in commit 1a5edef

Mode	Failure mode	Stage 4.5 round-1	Stage 4.5 round-2	Resolution
4	Shortcut reliance	PASS — RQ4 $\rho=0.16$ honestly framed as “not interpretable at $n=12$ ”	PASS	—
5	Bug-as-insight	PASS — v3b post-hoc selection explicitly declared as selection-on-the- response, not narrated as “MR design contribution”	PASS	—
6	Methodology fabrication	PASS — K_{eq} sweep honestly downgraded to limitation in round-2 Group B; not pretended to have been executed	PASS	—
7	Pipeline-level frame-lock	PASS — three-layer methodology backbone is primary, 60-cell empirical audit auxiliary; v3 / v3b / v4 alternative explanations not locked out	PASS	—

Root-cause analysis for Mode 1 + Mode 3: - Tip 2024 author hallucination: when adding references in round-1 Group C, the LLM relied on training-data co-occurrence frequency (Tip + Misailovic + Bavota) rather than actual paper author lists. **Mitigation:** any added reference must first WebFetch the DOI / arXiv ID to verify the author list. - DeepCrime fabrication: the LLM relied on narrative consistency (“DeepCrime + probabilistic programs”) rather than the actual paper subject (deep-learning systems). **Mitigation:** any subject-claim must WebFetch the paper abstract for confirmation. - cosmic-ray 1276 drift: when reporting in conversation the number was hand-typed as approximate rather than read from the SSOT. **Mitigation:** data claims must `grep data/results/*.json` for the value.

6 cumulative P0/P1 hallucination instances caught by the independent Stage 4.5 verifier — this demonstrates the value of academic-pipeline v3.2’s BLOCKING design. Without an independent verifier, these issues would silently enter the submitted manuscript.

4. Key lessons learned

1. **Cite-then-verify must be upfront:** round-1 added references without DOI verification → round-2 verifier caught fabrications. **Future workflow:** every in-conversation reference addition must be preceded by a WebFetch validation.
2. **Subagent-driven is a force multiplier:** the 5-reviewer parallel simulation + the subsequent independent verifier together are the central guarantee of paper quality; without an independent verifier, silent drops are unavoidable.
3. **R3 Major-Revision dissent should not have been overruled by 4-vs-1 majority:** R3’s §6.5 deployability concern was a substantive defect; the Group D 6-item fix was necessary, and the round-2 majority “all Minor” readiness was slightly premature. The spirit of academic-pipeline IRON RULE 4 (DA CRITICAL findings cannot be ignored) should extend to any high-quality critic dissent.

4. **Translation drift is a new risk class:** when □□□□P2_EN.md was generated via Anthropic API streaming, the §3.5.1 + §5.9.2-3 sections were dropped; round-2 ESCALATED caught this. **Future workflow:** every large-batch translation must verify section-count CN vs EN match.
 5. **Stage 4.5 final integrity check is non-redundant:** even with both round-1 and round-2 reviews PASS, the independent from-scratch verifier still found 3 P0 + 6 P1 issues.
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5. Pre-submission final checklist

- ☒ Manuscript: □□□□P2_EN.md (1844 lines, 5/5 reviewer Minor verdict)
 - ☒ Submission package: submission/p2_ist.{tex,pdf,docx}
 - ☒ Author block: Meng Li, mlemon@usc.edu.cn, USC-CN, 421001
 - ☒ Cover letter: submission/cover_letter.{md,pdf}
 - ☒ Stage 4.5 final integrity: PASS_WITH_NITS (0 P0)
 - ☒ Reproducibility: scripts/ + data/results/*.json + REPRODUCIBILITY.md
 - ☒ Audit trail: 30+ commits + docs/review_2026-05-01/ + docs/review_2026-05-02/
 - ☐ Zenodo DOI for replication package (deferred to user)
 - ☐ IST submission system upload (manual user step)
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